

Rapid Online Analysis of Photopolymerization Kinetics and Molecular Weight Using Diffusion NMR

Lucy L. Fillbrook, Mitchell D. Nothling, Martina H. Stenzel, William S. Price,* and Jonathon E. Beves*



Cite This: *ACS Macro Lett.* 2022, 11, 166–172



Read Online

ACCESS |



Metrics & More

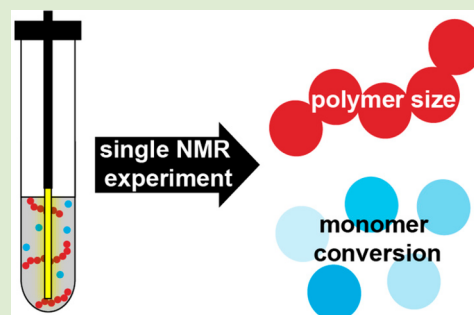


Article Recommendations



Supporting Information

ABSTRACT: Online, high-throughput molecular weight analysis of polymerizations is rare, with most studies relying on tedious sampling techniques and batchwise postanalysis. The ability to track both monomer conversion and molecular weight evolution in real time could underpin precision polymer development and facilitate study of rapid polymerization reactions. Here, we use a single time-resolved diffusion nuclear magnetic resonance (NMR) experiment to simultaneously study the kinetics and molecular weight evolution during a photopolymerization, with in situ irradiation inside the NMR instrument. As a model system, we used a photoinduced electron transfer reversible addition–fragmentation chain transfer (PET-RAFT) polymerization. The data allow diffusion coefficients and intensities to be calculated every 14 s from which the polymer size and monomer conversion can be extracted. Key to this approach is (1) the use of shuffled gradient amplitudes in the diffusion NMR experiment to access reactions of any rate, (2) the addition of a relaxation agent to increase achievable time resolution and, (3) a sliding correction that accounts for viscosity changes during polymerization. Diffusion NMR offers a uniquely simple, translatable handle for online monitoring of polymerization reactions.



Monitoring reactions in real time is vital for understanding reaction mechanisms and optimizing synthetic procedures. The achievable time-resolution and use of enclosed vessels enables the online analysis of rapid and highly sensitive processes, where batchwise processing is impractical. For polymerizations, online analysis of kinetics and molecular weight evolution generally requires flow reactors coupled with characterization techniques,¹ such as gel permeation chromatography (GPC).² The use of GPC limits studies to materials that fully dissolve in common solvents³ and confines the analysis to batchwise processing, restricting temporal resolution. Nuclear magnetic resonance (NMR) spectrometers are commonplace and are routinely used for kinetic and mechanistic studies.⁴ Inline benchtop NMR has even been used for rapid, automated screening of polymerizations.^{4d} Sometimes the ¹H NMR resonances of polymer chain and end group protons can be used to determine the apparent number-average molecular weight (M_n).^{4b,5} However, such analysis is fraught with difficulty when used in isolation³ and is not possible for large polymers or where the relevant NMR signals overlap. In such instances, diffusion NMR offers an alternative means of establishing polymer size⁶ and dispersity ($\bar{D}$).⁷

At infinite dilution, the self-diffusion coefficient (D) of any species is intrinsically linked to molecular size, shape, and solvent viscosity (η_0). For the simple case of a spherical particle with effective hydrodynamic radius r , the relationship is given by the Stokes–Einstein equation (“stick” boundary condition):⁸

$$D = \frac{k_B T}{6\pi\eta_0 r} \quad (1)$$

where k_B is the Boltzmann constant and T is the temperature. Using the Mark–Houwink relationship for monodisperse, infinitely dilute systems, the equation can be expressed in terms of the viscosity-average molecular weight (M_v):⁹

$$D = \frac{k_B T}{6\pi\eta_0} \left(\frac{10\pi N_A}{3KM_v^{\alpha+1}} \right)^{1/3} \quad (2)$$

where N_A is Avogadro’s number, and K and α are the Mark–Houwink coefficients specific to the polymer–solvent system. Simplifying eq 2, an empirical scaling relationship is obtained:^{10,6d}

$$D = kM_v^a \quad (3)$$

where k and a are variables specific to the polymer system that encompass the shape and density of the polymers in a given solvent and changes to solution viscosity (η) caused by polymer size. This relationship has been extended to the

Received: November 16, 2021

Accepted: January 5, 2022



weight-average molecular weight (M_w),¹¹ with calculated molecular weights comparing well with those from GPC for a range of polymer samples.^{6d,12}

Diffusion NMR measurements^{13a,6c,13b} are based on the pulsed gradient spin-echo (PGSE) NMR sequence and variations, such as the stimulated echo-based PGSTE sequence,^{6a-c} and can be used to obtain the diffusion coefficients of polymers. Sometimes the diffusion NMR data can be presented in a 2D display mode commonly referred to as diffusion-ordered spectroscopy (DOSY) to accentuate the ability of diffusion NMR to separate components in mixtures according to their diffusion coefficients.¹⁴ Diffusion NMR has been used to monitor the molecular weight evolution during polymerizations via batchwise sampling of the reaction mixture.^{12c} Online monitoring has been possible using a flow reactor¹⁵ or in a single NMR tube,¹⁶ though the methods were limited to polymerizations with rates that are slow compared to the diffusion NMR measurement time scale. In contrast, time-resolved diffusion NMR expands the range of reaction rates that can be studied by ensuring changes in concentration and relaxation rates do not impact the PGSE NMR measurement.^{17a,7e,17b,c} The method has been successfully applied to a variety of dynamic systems, including oligomer formation,¹⁸ polymer growth,^{17b} and polysaccharide degradation.^{17a}

Photoinduced electron transfer–reversible addition–fragmentation chain transfer (PET-RAFT) polymerization is a controlled radical process, now routinely used to prepare vinyl polymers with diverse architectures,¹⁹ such as the reaction shown in Figure 1a. The method is widely used due to excellent tolerance to oxygen, spatiotemporal control, and chain-end fidelity.¹⁹ Despite examples of online monitoring of kinetics using Fourier transform near-infrared²⁰ or NMR,²¹ no method exists for rapid online molecular weight analysis.

Herein we describe how time-resolved diffusion NMR experiments can be used in situ to monitor the kinetics and molecular size evolution during polymerizations. We demonstrate this approach using the PET-RAFT polymerization²² of methyl acrylate (MA) activated by zinc tetraphenylporphyrin (ZnTPP) photocatalyst as an example system (Figure 1a,b). Sample irradiation is achieved in situ using a fiber optic cable,²³ affording an online analysis technique that is broadly applicable to any polymerization with a low dispersity.

We aimed to simultaneously measure the monomer conversion and polymer molecular weight during a polymerization reaction using a single diffusion NMR experiment. In a PGSE sequence, the first magnetic gradient pulse (of duration δ and magnitude g) spatially encodes the transverse magnetization of each species into a helix. Diffusion in the direction of the gradient attenuates the helix. After a delay Δ , an identical but effectively negative gradient pulse is then applied to unwind the helix. Thus, Δ defines the time scale of the diffusion measurement. The acquired signal intensity (I) of each species will be attenuated by both spin relaxation over the duration of the sequence and by diffusion during Δ . Denoting the signal that would be acquired in the absence of diffusion (or equivalently, absence of a gradient pulse) as I_0 , the observed NMR signal intensity for a PGSE sequence is given by the Stejskal–Tanner equation (for a single freely diffusing species):^{13a,24}

$$I = I_0 \exp \left[-D(\gamma \delta g)^2 \left(\Delta - \frac{\delta}{3} \right) \right] = I_0 \exp[-Db] \quad (4)$$

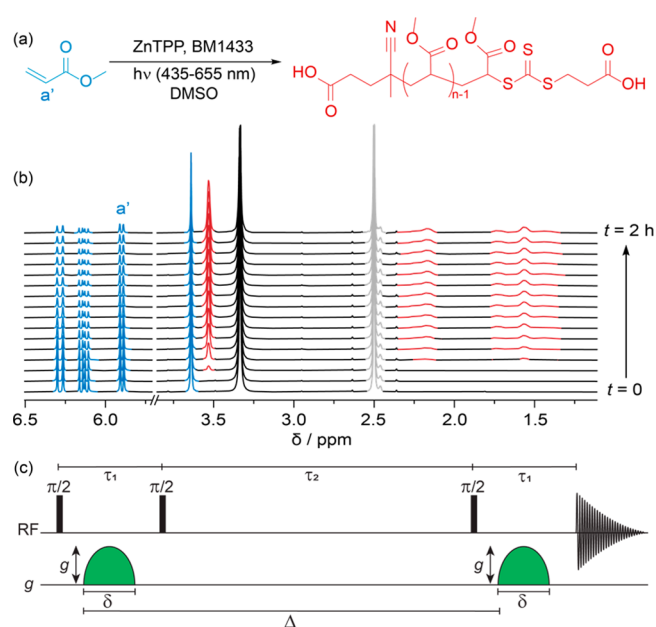


Figure 1. PET-RAFT polymerization of methyl acrylate (MA) to poly(methyl acrylate) (PMA) using photocatalyst zinc tetraphenylporphyrin (ZnTPP) and RAFT agent 4-(((2-carboxyethyl)thio)-carbonothioylthio)-4-cyanopentanoic acid (BM1433) in DMSO- d_6 .²² (a) Reaction scheme and (b) time-dependence of diffusion NMR spectra (^1H , 500 MHz, PGSTE, $\delta = 6$ ms, $g = 2.65$ G cm^{-1} , $\Delta = 50$ ms). Reaction conditions: $[\text{MA}]/[\text{BM1433}]/[\text{ZnTPP}]$: 100:1:0.01, 0.5 M MA, 170 μM Gd-DTPA, irradiation with 567 nm, 298 K. (c) The PGSTE pulse sequence used for diffusion NMR experiments, radiofrequency (RF) pulses and gradient pulses (g), defining time periods τ_1 and τ_2 , gradient amplitude g , gradient duration δ , and diffusion time Δ . The relaxation delay (not shown) is the time between the end of one application of the pulse sequence to the start of the next. See Figure S1 for full pulse sequence used in this work.

where γ is the gyromagnetic ratio. As shown, all the gradient parameters can be summarized by the diffusion weighting factor b . In the absence of spin relaxation changes, I_0 is proportional to the concentration of the species.

Normally a series of spectra are collected with b varied monotonically by altering g to achieve a signal attenuation of $\sim 90\%$ with the largest b value used (i.e., $bD \sim 2.3$).^{13b} Estimates for D and I_0 can then be obtained by nonlinear regression of eq 4 onto the data (Figure 2, right). However, for dynamic systems, g should be varied in a shuffled order to reduce any correlation between gradient ordering and nondiffusion-based intensity changes, that is, changes in concentrations or relaxation rates.^{17a,7e,17b} The use of a long shuffled list (e.g., 500 amplitudes) allows a moving fit (“windowing”) of eq 4 to be applied to a contiguously acquired set of spectra (Figure 2). This leads to improved time resolution that is crucial for studying rapid processes.^{7e,17b} In this work, a long shuffled list of 512 gradient amplitudes was created (SI-9), and diffusion data was processed in windows of 32 b values.

Parameter optimization for diffusion NMR experiments on static polymer samples has been discussed previously.²⁵ However, additional considerations are required for in situ measurements. For instance, it is useful to study both the monomer and the polymer during a polymerization, but the diffusion coefficients of these species may differ by orders of

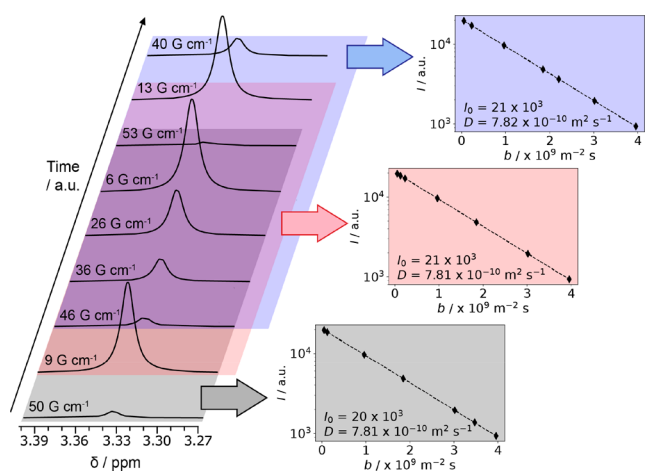


Figure 2. Principles of time-resolved diffusion NMR.^{7e,17b} Nine gradient slices from a diffusion NMR experiment acquired with shuffled g values during the PET-RAFT polymerization are displayed (NB only the water peak is shown). A moving fit to temporally contiguous subsets (three are shown) allows improved time-resolution of the estimated D and I_0 . Shuffling g values removes the correlation between the gradient order and nondiffusion-based intensity changes in dynamic systems.

magnitude due to the vastly different molecular sizes. Thus, $\sim 90\%$ attenuation will be unattainable for both species over the same range of gradient amplitudes. To tackle this issue, δ and Δ were chosen such that approximately half of the gradient amplitudes in a processing window (e.g., 4 of 7 gradient amplitudes in Figure 2) resulted in a measurable signal for the fast-diffusing species, while the remaining gradient amplitudes resulted in full attenuation and were excluded from processing. The shuffled gradient amplitude list was designed to ensure the diffusion coefficients of the fast-diffusing species could be accurately determined at all times. The signal for the largest polymer species was attenuated by $\sim 70\%$, but additional experiments with parameters for optimized attenuation ($bD \sim 2$, for the largest b) confirmed the diffusion coefficient was accurately estimated (SI-4.1).

The polymerization kinetics can be simultaneously measured directly from the calculated intensities (I_0), as long as I_0 is not affected by time-dependent spin-relaxation constants. To avoid interference from such relaxation changes during a reaction, Δ and the relaxation delay must be carefully selected. In the PGSTE NMR experiment, signal is lost to transverse relaxation (time constant T_2) during τ_1 and longitudinal relaxation (time constant T_1) during the τ_2 period (Figure 1c).^{26,8b} Thus, as Δ encompasses τ_2 , we require conditions of $\Delta < T_1$ for the proton with the shortest T_1 to avoid signal loss. Additionally, the relaxation delay should be $5 \times T_1$ for the proton with the longest T_1 to ensure sufficient relaxation and maximum signal of all protons, simplifying analysis. If a T_1 is long, the required experiment time may be impractical for studying fast processes. To improve time-resolution, a paramagnetic relaxation agent like gadolinium-diethylenetriamine pentaacetic acid (Gd-DTPA) can be added to reduce the T_1 of all protons in solution, allowing shorter relaxation delays,²⁷ as we demonstrate here.

We now consider the PET-RAFT polymerization of MA in Figure 1a. The longest T_1 of interest in the reaction solution prior to irradiation is 5.20 ± 0.01 s for the vinyl MA signal marked a' in Figure 1 (SI-5.1). Adding $170 \mu\text{M}$ Gd-DTPA

reduced this T_1 to 1.94 ± 0.05 s, with no change to the diffusion coefficient (SI-4.2). As such, the relaxation delay was set to 10 s ($5 \times T_1$), allowing one gradient slice to be collected every 14 s (~ 7 min to collect 32 gradient slices for the first data point). Similarly, Δ was set to 50 ms after measuring the shortest T_1 of the polymer (formed after 2 h) as 600 ± 10 ms (i.e., $T_1 \gg 50$ ms; SI-5.1). Time-resolved T_1 measurements (SI-1.3.3) during the PET-RAFT reaction confirmed these parameters would be suitable throughout the polymerization (SI-5.2).

With the diffusion NMR experiment optimized for the system, the polymerization was carried out in a NMR tube, using a fiber optic for continuous irradiation in the spectrometer.²³ To disrupt the convective flow known to result from heating during in situ irradiation,¹⁸ we used physical obstruction²⁸ by packing the NMR tube with glass capillaries (see SI-2.2) to ensure accuracy of the diffusion coefficients. The measurement was repeated three times to assess reproducibility. The resulting diffusion coefficients and intensity data obtained during the PET-RAFT polymerization are shown in Figure 3a,b. The intensity data (I_0) in Figure 3b are representative of species concentration due to the careful selection of relaxation delay and diffusion time discussed above (see SI-6.1 for full details). Thus, monomer conversion can be calculated during the reaction using the diffusion NMR data (Figure 3c). The final conversion of $66 \pm 1\%$ after 2 h of irradiation was verified with the integrated signal intensity from a standard ^1H NMR experiment (dashed line, SI-6.2). This monomer conversion is lower than the 83% conversion from a previous report,²⁹ which used similar experimental conditions. The reduced value may result from the use of capillaries to restrict transport rather than mixing the sample.

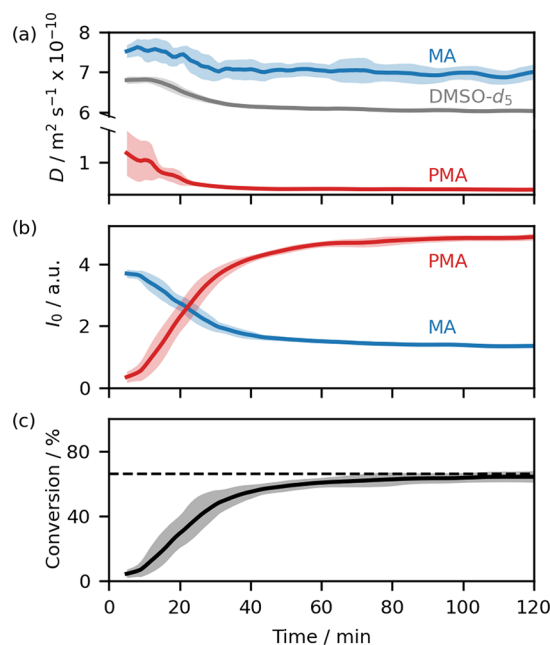


Figure 3. Simultaneous (a) diffusion, (b) intensity (I_0), and (c) conversion results from single diffusion NMR experiment (^1H , 500 MHz, PGSTE $\delta = 6$ ms, $\Delta = 50$ ms, $g = 3\text{--}50$ G cm^{-1} , 298 K) during PET-RAFT polymerization with in situ irradiation. Dashed line in (c) corresponds to conversion measured with ^1H NMR spectrum obtained after 2 h of irradiation. Shaded areas indicate the standard deviation over three experiments.

The diffusion coefficient of PMA after approximately 7 min irradiation was $1.2 \pm 0.6 \times 10^{-10} \text{ m}^2 \text{ s}^{-1}$ and decreased to $3.2 \pm 0.1 \times 10^{-11} \text{ m}^2 \text{ s}^{-1}$ after 2 h.³⁰ The decrease is caused by both the increasing polymer size and the increase in solution viscosity. The viscosity change is also evident in the decreased diffusion coefficients of the monomer and residual solvent (DMSO- d_6). Both species experience an approximate 10% decrease in their respective diffusion coefficients over the 2 h experiment, consistent with a 10% increase in solution viscosity.

The diffusion coefficient of a polymer can be used to estimate its molecular weight using eq 3. A suitable calibration curve should be constructed using standards with known molecular weights.^{6d,12c,16b,31,32} We prepared a calibration curve using poly(ethylene glycol) (PEG) standards^{12c} with peak molecular weight (M_p) of 1900–11850 g mol^{-1} (Figure 4), verified by GPC (SI-7.1). While eq 3 holds for infinitely

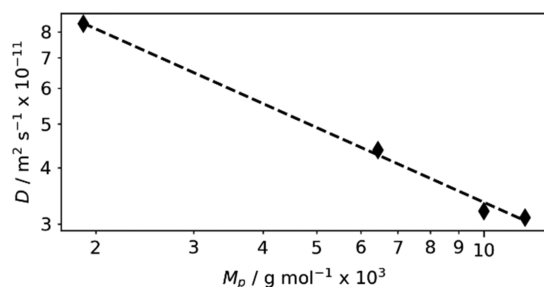


Figure 4. Calibration curve of PEG diffusing in DMSO- d_6 for estimating the molecular weight of a polymer. M_p verified with GPC and D from diffusion NMR experiments (^1H , 500 MHz, PGSTE, $\delta = 6 \text{ ms}$, $\Delta = 50 \text{ ms}$, $g = 3\text{--}50 \text{ G cm}^{-1}$, 298 K). Dashed line corresponds to nonlinear regression of eq 3 onto the data.

dilute systems,³³ the concentration of the polymer is high during the studied reaction, leading to significant interactions between molecules in solution and, thus, a higher solution viscosity. Hence, it is necessary to correct the diffusion coefficient for the observed viscosity change. From eq 1, the diffusion coefficient of each species will be equally affected by viscosity ($D \propto \eta^{-1}$), so the change in the diffusion coefficient of the solvent can be used to correct the diffusion coefficient of the polymer (SI-7.2).

The apparent molecular weight of the PMA polymer was estimated during the reaction for both the raw (D) and viscosity-corrected (D^*) diffusion coefficients (Figure 5a), expressed relative to the equivalent weight of a PEG polymer. After 2 h of irradiation, the apparent molecular weight calculated from the diffusion coefficient was $11000 \pm 600 \text{ g mol}^{-1}$, which decreased to $8300 \pm 500 \text{ g mol}^{-1}$ using the viscosity corrected diffusion coefficient.

The standards used to create the calibration curve had a $\bar{D} < 1.05$, so the molecular weight calculated using these standards is only accurate for samples with a similar dispersity. We used GPC to measure $\bar{D}$ of 1.3 after the reaction, with M_n of $8800 \pm 200 \text{ g mol}^{-1}$ and M_w of $11500 \pm 200 \text{ g mol}^{-1}$ (SI-7.3). Despite the relatively high dispersity of the reaction samples, the diffusion coefficient provides a good estimate for the molecular weight and, importantly, allows the molecular weight evolution to be tracked during the polymerization.

The molecular weight-conversion plot in Figure 5b shows a linear relationship, indicative of a living polymerization.³⁴ The plot indicates the usefulness of the outlined diffusion NMR

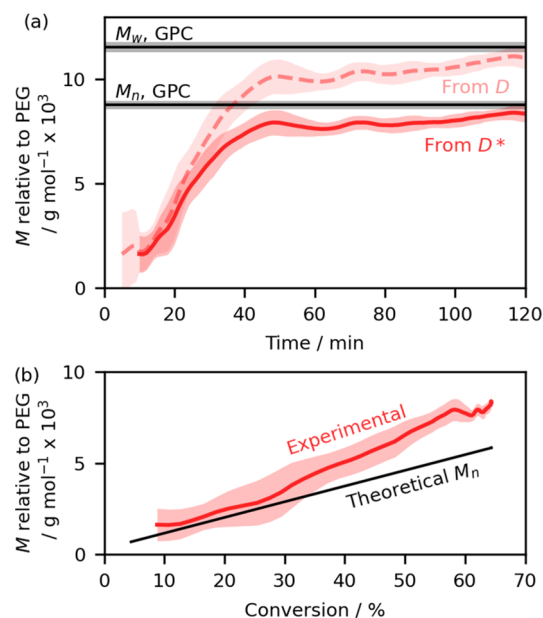


Figure 5. Molecular weight evolution during PET-RAFT polymerization. (a) Molecular weight calculated from time-resolved diffusion coefficient (raw D and viscosity-corrected D^*) using the calibration in Figure 4. The horizontal lines correspond to the M_n and M_w from GPC on the same sample after 2 h of irradiation. (b) Molecular weight-conversion plot. The solid black line indicates the theoretical M_n calculated by $M_{n(\text{theoretical})} = ([\text{MA}]_0/[\text{BM1433}]_0) \times (\%C) \times (MW_{\text{MA}} + MW_{\text{BM1433}})$, where $[\text{MA}]_0/[\text{BM1433}]_0$ = initial ratio of monomer to RAFT agent; $\%C$ = conversion = $I_{\text{polymer}}/(I_{\text{monomer}} + I_{\text{polymer}})$, I_n = intensity of ^1H NMR signal for n ; MW_{MA} and MW_{BM1433} = molecular weight of MA monomer and BM1433, respectively.³⁵ Experimental parameters: ^1H , 500 MHz, PGSTE $\delta = 6 \text{ ms}$, $\Delta = 50 \text{ ms}$, $g = 3\text{--}50 \text{ G cm}^{-1}$, 298 K. Shaded areas indicate the standard deviation averaged over three experiments.

experiment, with both molecular weight and conversion data extracted from the same experiment.

To demonstrate the versatility of the proposed method, we have also studied the aqueous RAFT polymerization of acrylic acid, GPC analysis of which is challenging³⁶ due to the charged systems created by the polyelectrolytes. With our method, the monomer conversion and molecular weight can also be estimated over time (see SI-8).

It is not possible to use the diffusion coefficient to measure dispersity with our method, although this has been previously demonstrated with diffusion NMR experiments.⁷ Using our diffusion NMR parameters, the NMR signal for largest polymer formed is attenuated by $\sim 70\%$ and the data are well-described by eq 4. To measure the dispersity, a deviation of the experimental data from the fit of eq 4 is required to model the distribution of diffusion coefficients. Such deviations are generally only detectable at very high attenuation.^{7g} Our methodology might be adapted to achieve these high levels of attenuation for the polymer, while also monitoring the faster diffusing species, by using a spectrometer capable of applying a larger maximum gradient amplitude (note in this work $g_{\text{max}} = 51 \text{ G cm}^{-1}$).

A single diffusion NMR experiment can be used to monitor polymerization kinetics and molecular weight evolution with high time-resolution (14 s between data points). This versatile method does not require any specialist equipment and can be applied to fast reaction processes in any solvent where NMR

spectra can be collected, including those that are unsuitable for GPC analysis. By eliminating the need for sequential or parallel analysis of properties, data can be efficiently collected to provide insights into highly time-sensitive polymerization reactions.

■ ASSOCIATED CONTENT

SI Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acsmacrolett.1c00719>.

Synthetic procedures, NMR experimental data, spectra, and analysis (PDF)

■ AUTHOR INFORMATION

Corresponding Authors

William S. Price – Nanoscale Group, School of Science, Western Sydney University, Penrith, NSW 2751, Australia; orcid.org/0000-0002-8549-4665; Email: w.price@westernsydney.edu.au

Jonathon E. Beves – School of Chemistry, UNSW Sydney, Sydney, NSW 2052, Australia; orcid.org/0000-0002-5997-6580; Email: j.beves@unsw.edu.au

Authors

Lucy L. Fillbrook – School of Chemistry, UNSW Sydney, Sydney, NSW 2052, Australia; orcid.org/0000-0001-9443-0834

Mitchell D. Nothling – School of Chemistry, UNSW Sydney, Sydney, NSW 2052, Australia; orcid.org/0000-0002-3259-0047

Martina H. Stenzel – School of Chemistry, UNSW Sydney, Sydney, NSW 2052, Australia; orcid.org/0000-0002-6433-4419

Complete contact information is available at: <https://pubs.acs.org/doi/10.1021/acsmacrolett.1c00719>

Author Contributions

The manuscript was written through contributions of all authors.

Funding

This work was supported by the Australian Research Council (FT170100094 to J.E.B.) and UNSW Australia (L.L.F.).

Notes

The authors declare no competing financial interest.

■ ACKNOWLEDGMENTS

We acknowledge the Mark Wainwright Analytical Centre at UNSW Sydney for access to the NMR facility.

■ REFERENCES

- (1) Haven, J. J.; Junkers, T. Online Monitoring of Polymerizations: Current Status. *Eur. J. Org. Chem.* **2017**, 2017, 6474–6482.
- (2) (a) Boussie, T. R.; Diamond, G. M.; Goh, C.; Hall, K. A.; LaPointe, A. M.; Leclerc, M.; Lund, C.; Murphy, V.; Shoemaker, J. A. W.; Tracht, U.; Turner, H.; Zhang, J.; Uno, T.; Rosen, R. K.; Stevens, J. C. A Fully Integrated High-Throughput Screening Methodology for the Discovery of New Polyolefin Catalysts: Discovery of a New Class of High Temperature Single-Site Group (IV) Copolymerization Catalysts. *J. Am. Chem. Soc.* **2003**, 125, 4306–4317. (b) Levere, M. E.; Willoughby, I.; O'Donohue, S.; Wright, P. M.; Grice, A. J.; Fidge, C.; Remzi Becer, C.; Haddleton, D. M. Cu(0) mediated polymerization in toluene using online rapid GPC monitoring. *J. Polym. Sci.* **2011**, 49, 1753–1763.
- (3) Stenzel, M. H.; Barner-Kowollik, C. The living dead – common misconceptions about reversible deactivation radical polymerization. *Mater. Horiz.* **2016**, 3, 471–477.
- (4) (a) Haddleton, D. M.; Perrier, S.; Bon, S. A. F. Copper(I)-Mediated Living Radical Polymerization in the Presence of Oxyethylene Groups: Online ^1H NMR Spectroscopy To Investigate Solvent Effects. *Macromolecules* **2000**, 33, 8246–8251. (b) Obermeier, B.; Wurm, F.; Frey, H. Amino Functional Poly(ethylene glycol) Copolymers via Protected Amino Glycidol. *Macromolecules* **2010**, 43, 2244–2251. (c) Knox, S. T.; Parkinson, S.; Stone, R.; Warren, N. J. Benchtop flow-NMR for rapid online monitoring of RAFT and free radical polymerisation in batch and continuous reactors. *Polym. Chem.* **2019**, 10, 4774–4778. (d) Rubens, M.; Van Herck, J.; Junkers, T. Automated Polymer Synthesis Platform for Integrated Conversion Targeting Based on Inline Benchtop NMR. *Macro Lett.* **2019**, 8, 1437–1441.
- (5) Izunobi, J. U.; Higginbotham, C. L. Polymer Molecular Weight Analysis by ^1H NMR Spectroscopy. *J. Chem. Educ.* **2011**, 88, 1098–1104.
- (6) (a) Callaghan, P. Pulsed Field Gradient Nuclear Magnetic Resonance as a Probe of Liquid State Molecular Organization. *Aust. J. Phys.* **1984**, 37, 359–388. (b) von Meerwall, E. D. *Spectroscopy: NMR, Fluorescence, FT-IR*; Springer: Berlin, Heidelberg, 1984; pp 1–29. (c) Stilbs, P. Fourier transform pulsed-gradient spin-echo studies of molecular diffusion. *Prog. Nucl. Magn. Reson. Spectrosc.* **1987**, 19, 1–45. (d) Chen, A.; Wu, D.; Johnson, C. S. Determination of Molecular Weight Distributions for Polymers by Diffusion-Ordered NMR. *J. Am. Chem. Soc.* **1995**, 117, 7965–7970. (e) Walderhaug, H.; Söderman, O.; Topgaard, D. Self-diffusion in polymer systems studied by magnetic field-gradient spin-echo NMR methods. *Prog. Nucl. Magn. Reson. Spectrosc.* **2010**, 56, 406–425. (f) Viéville, J.; Tanty, M.; Delsuc, M.-A. Polydispersity index of polymers revealed by DOSY NMR. *J. Magn. Reson.* **2011**, 212, 169–173. (g) Hiller, W. Quantitative Studies of Block Copolymers and Their Containing Homopolymer Components by Diffusion Ordered Spectroscopy. *Macromol. Chem. Phys.* **2019**, 220, 1900255.
- (7) (a) Håkansson, B.; Nydén, M.; Söderman, O. The influence of polymer molecular-weight distributions on pulsed field gradient nuclear magnetic resonance self-diffusion experiments. *Colloid Polym. Sci.* **2000**, 278, 399–405. (b) Gong, X.; Hansen, E. W.; Chen, Q. A Simple Access to the (Log/Normal) Molecular Weight Distribution Parameters of Polymers Using PGSTE NMR. *Macromol. Chem. Phys.* **2011**, 212, 1007–1015. (c) Röding, M.; Bernin, D.; Jonasson, J.; Särkkä, A.; Topgaard, D.; Rudemo, M.; Nydén, M. The gamma distribution model for pulsed-field gradient NMR studies of molecular-weight distributions of polymers. *J. Magn. Reson.* **2012**, 222, 105–111. (d) Röding, M.; Williamson, N. H.; Nydén, M. Gamma convolution models for self-diffusion coefficient distributions in PGSE NMR. *J. Magn. Reson.* **2015**, 261, 6–10. (e) Urbańczyk, M.; Bernin, D.; Czuroń, A.; Kazimierzczuk, K. Monitoring polydispersity by NMR diffusometry with tailored norm regularisation and moving-frame processing. *Analyst* **2016**, 141, 1745–1752. (f) Williamson, N. H.; Nydén, M.; Röding, M. The lognormal and gamma distribution models for estimating molecular weight distributions of polymers using PGSE NMR. *J. Magn. Reson.* **2016**, 267, 54–62. (g) Guo, X.; Laryea, E.; Wilhelm, M.; Luy, B.; Nirschl, H.; Guthausen, G. Diffusion in Polymer Solutions: Molecular Weight Distribution by PFG-NMR and Relation to SEC. *Macromol. Chem. Phys.* **2017**, 218, 1600440. (h) Williamson, N. H.; Röding, M.; Miklavcic, S. J.; Nydén, M. Scaling exponent and dispersity of polymers in solution by diffusion NMR. *J. Colloid Interface Sci.* **2017**, 493, 393–397.
- (8) (a) Cohen, Y.; Avram, L.; Frish, L. Diffusion NMR Spectroscopy in Supramolecular and Combinatorial Chemistry: An Old Parameter—New Insights. *Angew. Chem., Int. Ed.* **2005**, 44, 520–554. (b) Price, W. S. *NMR Studies of Translational Motion: Principles and Applications*; Cambridge University Press: Cambridge, 2009.
- (9) Rudin, A.; Johnston, H. K. Diffusion coefficients and intrinsic viscosities of polymers. *J. Polym. Sci. B Polym. Lett.* **1971**, 9, 55–60.

- (10) Flory, P. J. *Principles of Polymer Chemistry*; Cornell University Press: Ithaca, N.Y., 1953.
- (11) Raghavan, R.; Maver, T. L.; Blum, F. D. Nuclear magnetic resonance measurements of molecular weights. Self-diffusion of poly(methyl methacrylate) in acetonitrile. *Macromolecules* **1987**, *20*, 814–818.
- (12) (a) Mazarin, M.; Viel, S.; Allard-Breton, B.; Thévand, A.; Charles, L. Use of Pulsed Gradient Spin–Echo NMR as a Tool in MALDI Method Development for Polymer Molecular Weight Determination. *Anal. Chem.* **2006**, *78*, 2758–2764. (b) Barrère, C.; Mazarin, M.; Giordanengo, R.; Phan, T. N. T.; Thévand, A.; Viel, S.; Charles, L. Molecular Weight Determination of Block Copolymers by Pulsed Gradient Spin Echo NMR. *Anal. Chem.* **2009**, *81*, 8054–8060. (c) Li, W.; Chung, H.; Daeflfer, C.; Johnson, J. A.; Grubbs, R. H. Application of ^1H DOSY for Facile Measurement of Polymer Molecular Weights. *Macromolecules* **2012**, *45*, 9595–9603. (d) Chalmignon, C.; Duret, D.; Charreyre, M.-T.; Favier, A. ^1H DOSY NMR Determination of the Molecular Weight and the Solution Properties of Poly(N-acryloylmorpholine) in Various Solvents. *Macromol. Chem. Phys.* **2016**, *217*, 2286–2293. (e) Cherifi, N.; Khoukh, A.; Benaboura, A.; Billon, L. Diffusion-ordered spectroscopy NMR DOSY: an all-in-one tool to simultaneously follow side reactions, livingness and molar masses of polymethylmethacrylate by nitroxide mediated polymerization. *Polym. Chem.* **2016**, *7*, 5249–5257. (f) Gu, K.; Onorato, J.; Xiao, S. S.; Luscombe, C. K.; Loo, Y.-L. Determination of the Molecular Weight of Conjugated Polymers with Diffusion-Ordered NMR Spectroscopy. *Chem. Mater.* **2018**, *30*, 570–576. (g) Arrabal-Campos, F. M.; Aguilera-Sáez, L. M.; Fernández, I. Algebraic Reconstruction Technique for Diffusion NMR Experiments. Application to the Molecular Weight Prediction of Polymers. *J. Phys. Chem. A* **2019**, *123*, 943–950. (h) Zhang, C.; Jin, Z.; Zeng, B.; Wang, W.; Palui, G.; Mattoussi, H. Characterizing the Brownian Diffusion of Nanocolloids and Molecular Solutions: Diffusion-Ordered NMR Spectroscopy vs Dynamic Light Scattering. *J. Phys. Chem. B* **2020**, *124*, 4631–4650. (i) Hou, J.; Pearce, E. Characterization of Polymer Molecular Weight Distribution by NMR Diffusometry: Experimental Criteria and Findings. *Anal. Chem.* **2021**, *93*, 7958–7964. (j) Martínez, C. R.; Pérez, J. M.; Arrabal-Campos, F. M.; Batuecas, M.; Ortuño, M. A.; Fernández, I. Cyclic polylactide synthesis initiated by a lithium anthraquinoid: understanding the selectivity through DFT and diffusion NMR. *Polym. Chem.* **2021**, *12*, 4083–4092.
- (13) (a) Stejskal, E. O.; Tanner, J. E. Spin Diffusion Measurements: Spin Echoes in the Presence of a Time-Dependent Field Gradient. *J. Chem. Phys.* **1965**, *42*, 288–292. (b) Gupta, A.; Stait-Gardner, T.; Price, W. S. NMR imaging and diffusion. *Adsorption* **2021**, *27*, 503–533.
- (14) (a) Stilbs, P. Molecular self-diffusion coefficients in Fourier transform nuclear magnetic resonance spectrometric analysis of complex mixtures. *Anal. Chem.* **1981**, *53*, 2135–2137. (b) Morris, K. F.; Johnson, C. S. Diffusion-ordered two-dimensional nuclear magnetic resonance spectroscopy. *J. Am. Chem. Soc.* **1992**, *114*, 3139–3141.
- (15) Vrijssen, J. H.; Thomlinson, I. A.; Levere, M. E.; Lyall, C. L.; Davidson, M. G.; Hintermair, U.; Junkers, T. Online tracing of molecular weight evolution during radical polymerization via high-resolution FlowNMR spectroscopy. *Polym. Chem.* **2020**, *11*, 3546–3550.
- (16) (a) Price, W. S.; Tsuchiya, F.; Arata, Y. Time Dependence of Aggregation in Crystallizing Lysozyme Solutions Probed Using NMR Self-Diffusion Measurements. *Biophys. J.* **2001**, *80*, 1585–1590. (b) Lewinski, P.; Sosnowski, S.; Kazmierski, S.; Penczek, S. L-Lactide polymerization studied by ^1H NMR with diffusion-ordered spectroscopy (DOSY): a “One NMR Tube Experiment” providing data on monomer conversion, polymer structure, Mn and Mw. *Polym. Chem.* **2015**, *6*, 4353–4357.
- (17) (a) Oikonomou, M.; Asencio-Hernández, J.; Velders, A. H.; Delsuc, M.-A. Accurate DOSY measure for out-of-equilibrium systems using permuted DOSY (p-DOSY). *J. Magn. Reson.* **2015**, *258*, 12–16. (b) MacDonald, T. S. C.; Price, W. S.; Beves, J. E. Time-Resolved Diffusion NMR Measurements for Transient Processes. *ChemPhysChem* **2019**, *20*, 926–930. (c) Lee, A. M.; Stait-Gardner, T.; Price, W. S. Explicit phenomenological solutions for magnetization exposed to an arbitrary NMR diffusion steady state pulse sequence. *J. Chem. Phys.* **2021**, *155*, 144204.
- (18) MacDonald, T. S. C.; Feringa, B. L.; Price, W. S.; Wezenberg, S. J.; Beves, J. E. Controlled Diffusion of Photoswitchable Receptors by Binding Anti-electrostatic Hydrogen-Bonded Phosphate Oligomers. *J. Am. Chem. Soc.* **2020**, *142*, 20014–20020.
- (19) Nothling, M. D.; Fu, Q.; Reyhani, A.; Allison-Logan, S.; Jung, K.; Zhu, J.; Kamigaito, M.; Boyer, C.; Qiao, G. G. Progress and Perspectives Beyond Traditional RAFT Polymerization. *Adv. Sci.* **2020**, *7*, 2001656.
- (20) Wu, C.; Jung, K.; Ma, Y.; Liu, W.; Boyer, C. Unravelling an oxygen-mediated reductive quenching pathway for photopolymerisation under long wavelengths. *Nat. Commun.* **2021**, *12*, 478.
- (21) (a) Dolinski, N. D.; Page, Z. A.; Eisenreich, F.; Niu, J.; Hecht, S.; Read de Alaniz, J.; Hawker, C. J. A Versatile Approach for In Situ Monitoring of Photoswitches and Photopolymerizations. *ChemPhotoChem* **2017**, *1*, 125–131. (b) Niu, J.; Page, Z. A.; Dolinski, N. D.; Anastasaki, A.; Hsueh, A. T.; Soh, H. T.; Hawker, C. J. Rapid Visible Light-Mediated Controlled Aqueous Polymerization with In Situ Monitoring. *Macro Lett.* **2017**, *6*, 1109–1113.
- (22) Shanmugam, S.; Xu, J.; Boyer, C. Exploiting Metalloporphyrins for Selective Living Radical Polymerization Tunable over Visible Wavelengths. *J. Am. Chem. Soc.* **2015**, *137*, 9174–9185.
- (23) (a) Feldmeier, C.; Bartling, H.; Riedle, E.; Gschwind, R. M. LED based NMR illumination device for mechanistic studies on photochemical reactions – Versatile and simple, yet surprisingly powerful. *J. Magn. Reson.* **2013**, *232*, 39–44. (b) Mallo, N.; Brown, P. T.; Iranmanesh, H.; MacDonald, T. S. C.; Teusner, M. J.; Harper, J. B.; Ball, G. E.; Beves, J. E. Photochromic switching behaviour of donor–acceptor Stenhouse adducts in organic solvents. *Chem. Commun.* **2016**, *52*, 13576–13579. (c) Seegerer, A.; Nitschke, P.; Gschwind, R. M. Combined In Situ Illumination-NMR-UV/Vis Spectroscopy: A New Mechanistic Tool in Photochemistry. *Angew. Chem., Int. Ed.* **2018**, *57*, 7493–7497. (d) Ji, Y.; DiRocco, D. A.; Kind, J.; Thiele, C. M.; Gschwind, R. M.; Reibarkh, M. LED-Illuminated NMR Spectroscopy: A Practical Tool for Mechanistic Studies of Photochemical Reactions. *ChemPhotoChem* **2019**, *3*, 984–992. (e) Nitschke, P.; Lokesh, N.; Gschwind, R. M. Combination of illumination and high resolution NMR spectroscopy: Key features and practical aspects, photochemical applications, and new concepts. *Prog. Nucl. Magn. Reson. Spectrosc.* **2019**, *114–115*, 86–134.
- (24) The equation is modified slightly for the pulsed-field gradient stimulated echo (PGSTE) sequence that was used in this work.
- (25) (a) Price, W. S. Pulsed-field gradient nuclear magnetic resonance as a tool for studying translational diffusion: Part II. Experimental aspects. *Concepts Magn. Reson.* **1998**, *10*, 197–237. (b) Groves, P. Diffusion ordered spectroscopy (DOSY) as applied to polymers. *Polym. Chem.* **2017**, *8*, 6700–6708.
- (26) As molecules get larger, T2 decreases much more than T1 (for very large molecules, T1 begins to increase while T2 continues to decrease). Thus, even though $\tau_1 \ll \tau_2$, a lot of signal can be lost during τ_1 due to T2.
- (27) Strich, G.; Hagan, P. L.; Gerber, K. H.; Slutsky, R. A. Tissue distribution and magnetic resonance spin lattice relaxation effects of gadolinium-DTPA. *Radiology* **1985**, *154*, 723–726.
- (28) (a) Price, W. S.; Ide, H.; Arata, Y. Translational and rotational motion of isolated water molecules in nitromethane studied using ^{17}O NMR. *J. Chem. Phys.* **2000**, *113*, 3686–3689. (b) Swan, I.; Reid, M.; Howe, P. W. A.; Connell, M. A.; Nilsson, M.; Moore, M. A.; Morris, G. A. Sample convection in liquid-state NMR: Why it is always with us, and what we can do about it. *J. Magn. Reson.* **2015**, *252*, 120–129.
- (29) Dolinski, N. D.; Page, Z. A.; Discekici, E. H.; Meis, D.; Lee, I.-H.; Jones, G. R.; Whitfield, R.; Pan, X.; McCarthy, B. G.; Shanmugam, S.; Kottisch, V.; Fors, B. P.; Boyer, C.; Miyake, G. M.; Matyjaszewski, K.; Haddleton, D. M.; Alaniz, J. R.; Anastasaki, A.; Hawker, C. J. What happens in the dark? Assessing the temporal control of photo-

mediated controlled radical polymerizations. *J. Polym. Sci.* **2019**, *57*, 268–273.

(30) The system is crowded as the average spacing between molecules is much smaller than the mean-squared displacement from the Einstein equation for the time scale of the experiment. This means the apparent diffusion coefficients may appear smaller than the true diffusion coefficient due to obstruction. See reference for more details: Price, W. S.; Tsuchiya, F.; Arata, Y. Lysozyme Aggregation and Solution Properties Studied Using PGSE NMR Diffusion Measurements. *J. Am. Chem. Soc.* **1999**, *121*, 11503–11512.

(31) (a) Arrabal-Campos, F. M.; Oña-Burgos, P.; Fernández, I. Molecular weight prediction with no dependence on solvent viscosity. A quantitative pulse field gradient diffusion NMR approach. *Polym. Chem.* **2016**, *7*, 4326–4329. (b) Zaccaria, F.; Zuccaccia, C.; Cipullo, R.; Macchioni, A. Extraction of Reliable Molecular Information from Diffusion NMR Spectroscopy: Hydrodynamic Volume or Molecular Mass? *Chem.-Eur. J.* **2019**, *25*, 9930–9937.

(32) Voorter, P.-J.; McKay, A.; Dai, J.; Paravagna, O.; Cameron, N. R.; Junkers, T. Solvent-Independent Molecular Weight Determination of Polymers Based on a Truly Universal Calibration. *Angew. Chem., Int. Ed.* **2021**, DOI: 10.1002/anie.202114536.

(33) Equation 3 can be modified to account for higher concentrations. See ref 7g.

(34) Braunecker, W. A.; Matyjaszewski, K. Controlled/living radical polymerization: Features, developments, and perspectives. *Prog. Polym. Sci.* **2007**, *32*, 93–146.

(35) Perrier, S. 50th Anniversary Perspective: RAFT Polymerization—A User Guide. *Macromolecules* **2017**, *50*, 7433–7447.

(36) (a) Lacík, I.; Beuermann, S.; Buback, M. PLP–SEC Study into Free-Radical Propagation Rate of Nonionized Acrylic Acid in Aqueous Solution. *Macromolecules* **2003**, *36*, 9355–9363. (b) Lacík, I.; Stach, M.; Kasák, P.; Semak, V.; Uhelská, L.; Chovancová, A.; Reinhold, G.; Kilz, P.; Delaittre, G.; Charleux, B.; Chaduc, I.; D'Agosto, F.; Lansalot, M.; Gaborieau, M.; Castignolles, P.; Gilbert, R. G.; Szablan, Z.; Barner-Kowollik, C.; Hesse, P.; Buback, M. SEC Analysis of Poly(Acrylic Acid) and Poly(Methacrylic Acid). *Macromol. Chem. Phys.* **2015**, *216*, 23–37.



ACS IN FOCUS

Cellulose
Lab-Grown
Dilek Erilliç
Dorothee E.

Machine Learning in Chemistry
Jon Paul Janet & Heather J. Kulik

bacterials
Joria Cheng Jaramillo
William M. Wuest

ACS Publications

ACS In Focus ebooks are digital publications that help readers of all levels accelerate their fundamental understanding of emerging topics and techniques from across the sciences.

pubs.acs.org/series/infocus

ACS Publications
Most Trusted. Most Cited. Most Read.